File permissions in Linux

Project description

This project focuses on managing file and directory permissions in a Linux environment to enforce proper authorization. It involves reviewing existing permissions within the `/home/researcher2/projects` directory, identifying any misconfigurations, and applying appropriate changes to ensure only authorized users and groups have access.

Check file and directory details

First, I must explore the permissions of the projects directory and the files it contains. The lab starts with `/home/researcher2` as the current working directory. As seen in the command-line interface below, I used the cd command to enter the projects directory and check for the permissions for all files in the directory, including any hidden files.

```
researcher2@e624d0fc8b83:~$ cd projects
researcher2@e624d0fc8b83:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Apr 26 06:49 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Apr 26 06:49 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Apr 26 06:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_t.txt
researcher2@e624d0fc8b83:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 26 06:49 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 26 08:29 ..
-rw--w---- 1 researcher2 research_team   46 Apr 26 06:49 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 26 06:49 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Apr 26 06:49 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Apr 26 06:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_t.txt
```

Describe the permissions string

“`drwx--x--- 2 researcher2 research_team 4096 Apr 26 06:49 drafts`”

A 10-character string begins each entry and indicates how the permissions on the file are set. A directory with full permissions for all owner types would be `drwxrwxrwx`:

For the line chosen above, the 10-character string `drwx--x---` represents the file type and permissions for the drafts directory. The first character d indicates that it is a directory. The next three characters `rwx` show that the user (owner) has full permissions: read, write, and execute. The following three characters `--x` indicate that the group has only execute permission, meaning they can access the directory but cannot read or modify its contents. The final three characters `---` show that all other users have no permissions, so they cannot access the directory at all.

Change file permissions

As indicated by the task, none of the files should allow the other users to write to files.
As shown by the output earlier, the `project_k.txt` file has write permissions for other users.

I changed the permissions of the file identified in the previous step so that the owner type of other doesn't have write permissions. The command-line interface shows the command I entered as well as the resulting change in permissions for the `project_k.txt` file.

```
researcher2@e624d0fc8b83:~/projects$ chmod o-w project_k.txt
researcher2@e624d0fc8b83:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Apr 26 06:49 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Apr 26 06:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_t.txt
```

Next, the task indicates that the file `project_m.txt` is a restricted file and should not be readable or writable by the group or other; only the user should have these permissions on this file. As listed earlier, it can be seen that the group permissions of the `project_m.txt` file is read only.

I used the chmod command to change permissions of the project_m.txt file so that the group doesn't have read or write permissions. The command-line interface shows the command I entered as well as the resulting change in permissions for the `project_m.txt` file.

```
researcher2@e624d0fc8b83:~/projects$ chmod g-r project_m.txt
researcher2@e624d0fc8b83:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Apr 26 06:49 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_k.txt
-rw------- 1 researcher2 research_team   46 Apr 26 06:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_t.txt
```

Change file permissions on a hidden file

The tasks indicated that the file `.project_x.txt` is a hidden file that has been archived and should not be written to by anyone. (The user and group should still be able to read this file.) It can be seen in the output earlier when listing the permissions for all files including hidden files that the user and group owner types have incorrect write permissions.

I changed the permissions of the file `.project_x.txt` using the chmod command so that both the user and the group can read, but not write to, the file. The command-line interface shows the command I entered as well as the resulting change in permissions for the `.project_x.txt` file.

```
researcher2@e624d0fc8b83:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@e624d0fc8b83:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 26 06:49 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 26 08:29 ..
-r--r----- 1 researcher2 research_team   46 Apr 26 06:49 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 26 06:49 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_k.txt
-rw------- 1 researcher2 research_team   46 Apr 26 06:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_t.txt
researcher2@e624d0fc8b83:~/projects$
```

Change directory permissions

The task indicated that only the `researcher2` user should be allowed to access the drafts directory and its contents. (This means that only researcher2 should have execute privileges.) It can be seen in the output earlier when exploring the permissions that the group has execute permissions and therefore has access to the drafts directory.

I removed the execute permission for the group from the drafts directory. The command-line interface shows the command I entered as well as the resulting change in permissions for the `drafts` directory.

```
researcher2@e624d0fc8b83:~/projects$ chmod g-x drafts
researcher2@e624d0fc8b83:~/projects$ ls -l
total 20
drwx------ 2 researcher2 research_team 4096 Apr 26 06:49 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_k.txt
-rw------- 1 researcher2 research_team   46 Apr 26 06:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 26 06:49 project_t.txt
researcher2@e624d0fc8b83:~/projects$
```

Summary

In this project, Linux file and directory permissions were reviewed and modified to enforce proper authorization within the `/home/researcher2/projects` directory. Incorrect permissions were identified and corrected using commands like chmod, ensuring that only the appropriate user (`researcher2`) has the required access while restricting group and other users from unauthorized reading, writing, or executing.